Linolium: A Containerized Workflow for Automated Pathogen Lineage Designation and Interactive Curation

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Abstract:

The rapid increase in pathogen genomic data has challenged traditional methods of lineage designation which often rely on manual inspection and curation by experts. While tools exist for lineage inference and phylogenetic visualization, they often operate in isolation, requiring researchers to navigate fragmented workflows and manual data transformations. To address this, we developed Linolium, an end-to-end, containerized workflow that integrates automated lineage discovery with interactive curation. By orchestrating AutoLin, TaxoniumTools, and Taxonium within a single Docker environment, Linolium automates the transition from Mutation Annotated Tree (MAT) data to browser-based review. We extended the utility of these tools by introducing metadata-weighted scoring, which allows users to prioritize lineages based on clinical or geographic context. Additionally, a new lineage curation interface enables real-time, node-level reassignment and data export, effectively bridging the gap between automated proposals and expert-driven revision. Linolium provides a scalable, user-friendly solution for the global surveillance of pathogens such as Tuberculosis, Mpox, and SARS-CoV-2, significantly lowering the technical barriers to reproducible genomic analysis.

Summary:

In public health, pathogen lineage designation is the process of identifying distinct groups within a species which is essential for tracking the transmission of infections and informing treatment strategies[1,2]. Linolium is an end-to-end containerized workflow for assigning and exploring lineage designations for pathogens using genomic data. It orchestrates lineage inference, metadata integration, and browser-based review within a single reproducible environment. Linolium enables users to assign, inspect, and refine lineage systems using genomic data represented in Mutation Annotated Tree (MAT) format[3], supporting both established nomenclatures and the development of new lineage systems.

Statement of Need:

Recent surges in pathogen sequence availability have challenged existing methods of lineage detection and curation. Pathogen surveillance methods developed during the SARS-CoV-2 pandemic, including wastewater-based tracking[4,5], have catalyzed a transition toward high-volume sequencing. Yet, this abundance of data has exposed a bottleneck in our ability to maintain up to date lineages and identify emerging threats, necessitating a more standardized and scalable approach to pathogen genomics. Existing tools address lineage identification and phylogenetic visualization separately, requiring users to manually connect outputs across workflows.

While our research group previously developed individual tools for these tasks[1,3,6–8], users must still navigate fragmented processes to achieve a full analysis. Linolium addresses this gap by automating the multi-tool requirements of lineage designation, as well as contributing specialized visualization and data handling into a streamlined environment. Designed for public health researchers, this tool reduces the manual coordination typically required for large pathogen datasets by integrating inference and curation steps. Linolium combines existing lineage designation and visualization tools and introduces metadata-aware lineage scoring within a unified workflow. The tool lowers technical barriers by encapsulating dependencies and managing intermediate data automatically.

State of the field:

Pathogen lineage designation is a decentralized process across many pathogen-specific nomenclatures[2,9–17] and a manual suggestion and curation process which is time-consuming and biased by human opinion[1,18]. In response to pandemic level sequencing, new tools such as AutoLin and Taxonium were developed which assist with elements of the lineage identification process such as candidate lineage identification and phylogenetic visualization. Despite the development of these tools, there was no tool which supported the identification of new lineages from beginning to end. Linolium orchestrates AutoLin[1], TaxoniumTools[7], and Taxonium[6], managing dependencies and intermediate data while extending AutoLin with metadata-weighted lineage scoring and integrating interactive curation. Linolium consolidates existing lineage designation and visualization tools into a unified workflow for automated and interactive lineage curation.

AutoLin is a pathogen-agnostic software tool which assists in automated, efficient, and phylogenetic-based lineage designation for faster identification of differentiation within species[1]. While AutoLin is highly versatile and scalable across various pathogens, it lacks a function to weight phenotypes defined in metadata which would provide more biologically informed lineage suggestions. Linolium adds additional parameters to AutoLin that allow the user to improve the biological relevance of the proposed lineages (see Research Impact).

Taxonium is a web-based, interactive visualization tool for large (>10M samples) phylogenetic trees and associated metadata. The original Taxonium tool is capable of displaying lineage information for nodes within a phylogeny but lacks the ability to identify AutoLin suggested lineages and curate this information interactively. The Linolium implementation of Taxonium has a new user interface that specifically identifies AutoLin outputs and allows for immediate review of the proposed new lineages.

Conversion of data across this analysis requires TaxoniumTools, a useful but technical tool that translates phylogenetic tree files into Taxonium readable files. Because TaxoniumTools has many features and specific use cases that require advanced knowledge and significantly increase technical overhead, Linolium replaces user interaction with this tool, handling all necessary inputs and outputs to convert informational results into interpretable visualizations with no additional commands.

Software Design:

Linolium is a containerized workflow that connects analysis, conversion, and visualization tools. While AutoLin and Taxonium are effective independently, their lack of integration necessitates significant manual data transformation and multiple command-line steps. We integrated these tools within a Docker container to ensure reproducibility and simplify dependency management across heterogeneous computing environments common in public health genomics.

Rather than developing competing software, we chose to build upon existing open-source tools to improve their efficiency and biological relevance. Linolium uses data formats and tools that are well documented and commonly used to build on existing frameworks and provide reliable results. All of its analyses and visualizations are built around the UShER Mutation Annotated Tree (MAT) format[3], which has become widely used in public health applications, tracking infectious disease evolution[19–23]. We extended the core logic of AutoLin by introducing a metadata-weighting mechanism that incorporates user-defined contextual variables into lineage scoring and prioritization. This allows the software to accept user-defined metadata, such as geographic or clinical importance, ensuring that the resulting lineage assignments are tailored to the specific biological context of the research.

The Linolium lineage curation browser interface is built with the Taxonium library. We extended Taxonium with lineage-aware visualization components that directly ingest AutoLin proposals (Figure 1), enable node-level lineage reassignment (Figure 2 and 3), and export updated lineage annotations in a format compatible with downstream phylogenetic analysis (Figure 1). This creates a unified environment in which automated lineage inference and expert-driven revision operate on the same underlying tree representation.. This design supports iterative cycles of automated lineage proposal followed by expert curation within the same analytical environment. By centralizing these tools into a unified, containerized application, Linolium consolidates previously separate command-line workflows into a single reproducible system for lineage designation and curation.

Figure 1: Visualization of M. tuberculosis Lineage 4.8 phylogeny with pre-existing and AutoLin assigned lineages within Linolium browser interface.

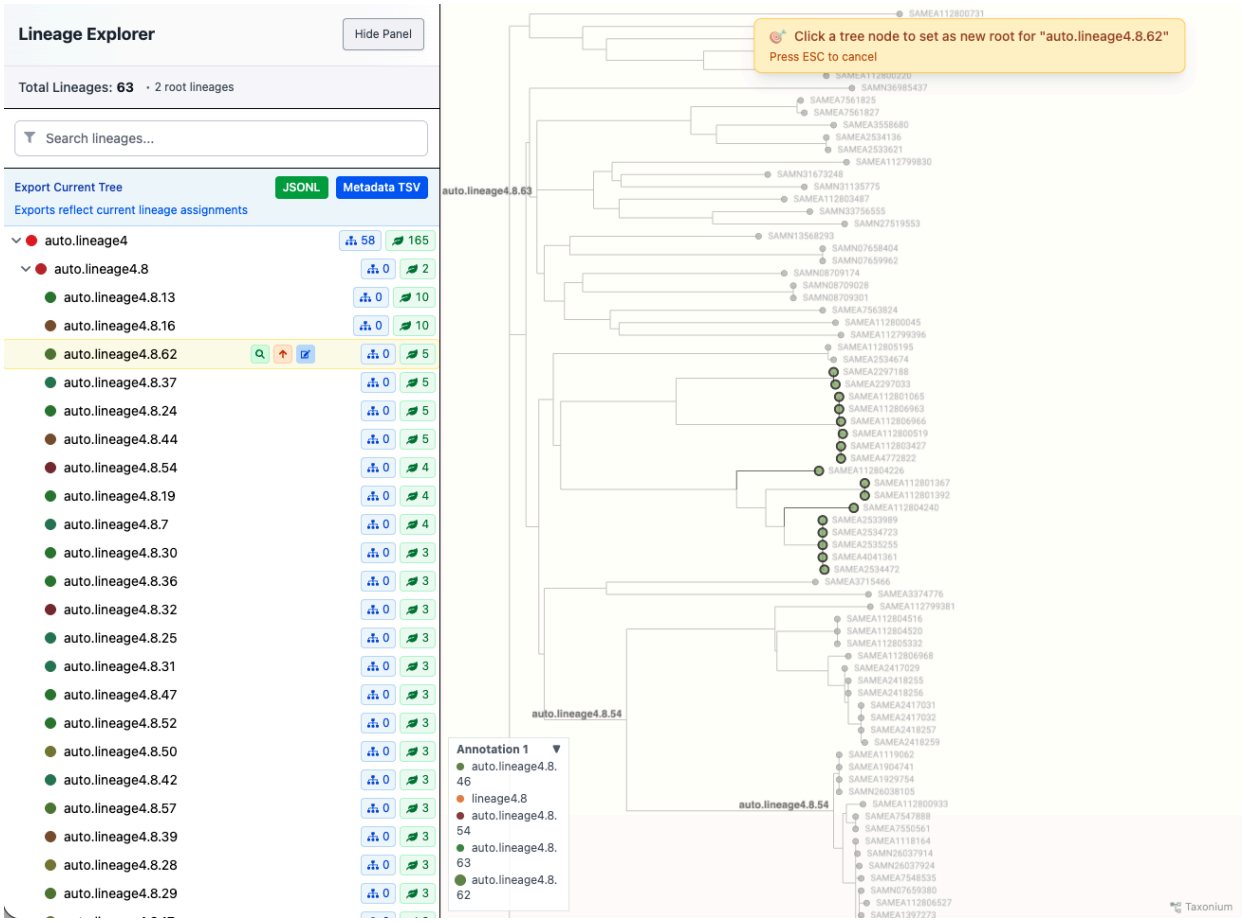

Figure 2: Linolium browser in lineage reassignment node for AutoLin designated lineage 4.8.62. In this mode any node can be selected as the new parent node for the lineage. In addition to parent node reassignment, a lineage can also be merged into the parent lineage, essentially rejecting a suggested lineage.

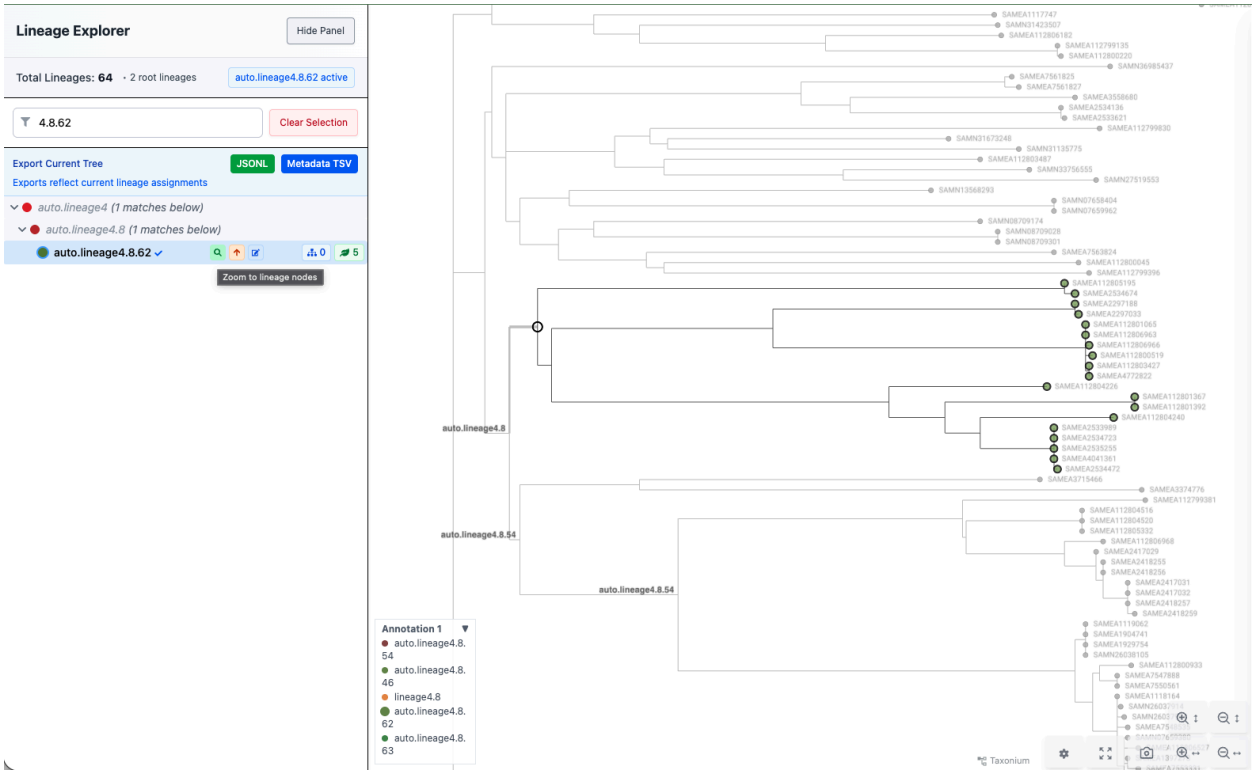

Figure 3: AutoLin designated Lineage auto.lineage4.8.62 after parent node reassignment (see Figure 2). Node reassignment will identify all descendent leaves and assign their correct lineage.

Because the ultimate goal of the analysis is interactive exploration, we chose to host all user interaction within a locally-hosted browser-based interface within the Docker container. This design removes the requirement for command-line expertise, allowing users to upload data and manage parameters through a graphical environment. This integrated approach transforms a series of isolated technical tasks into a complete, accessible pathway for pathogen surveillance.

Research Impact Statement:

To ensure immediate utility for public health genomic surveillance, the Linolium pipeline is built as a high-performance, reproducible environment that is currently ready for external adoption. A suite of Python command line tools powers the backend, while a locally hosted web interface manages the workflow, enabling users to execute and monitor analysis pipelines sequentially. The web interface, lineage designation tools, and interactive visualization are fully containerized using Docker, allowing for deployment on many computing environments without the need for manual dependency management. Community-readiness is further signaled by the public availability of the source code at https://github.com/corbett-lab/lineage-curation/tree/main, which includes highly documented README and example datasets to facilitate rapid onboarding for new users. By leveraging the UShER framework, Linolium is natively compatible with existing global-scale Mutation Annotated Trees (MATs), making it an immediately viable solution for

researchers monitoring Tuberculosis, Monkey Pox, Dengue fever and many many other pathogens.

Data Availability:

Complete documentation and example datasets found at https://github.com/corbett-lab/lineage-curation/tree/main.

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Citations:

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